```python
def is_palindrome(s):
    return s == s[::-1]


def first_palindrome(words):
    for word in words:
        if is_palindrome(word):
            return word

    return ""


words = ["abc", "car", "ada", "racecar", "cool"]

print(first_palindrome(words))



def find_intersection_values(nums1, nums2):
    set1 = set(nums1)
    set2 = set(nums2)

    answer1 = 0
    answer2 = 0

    for x in nums1:
        if x in set2:
            answer1 += 1

    for x in nums2:
        if x in set1:
            answer2 += 1

    return [answer1, answer2]


nums1 = [2, 3, 2]
nums2 = [1, 2]

print(find_intersection_values(nums1, nums2))


def sum_counts(nums):
    n = len(nums)
    answer = 0

    for i in range(n):
        distinct = set()

        for j in range(i, n):
            distinct.add(nums[j])

            count = len(distinct)
            answer += count * count

    return answer


nums = [1, 2, 1]

print(sum_counts(nums))


def count_pairs(nums, k):
    positions = {}
    answer = 0

    for j in range(len(nums)):
```

```
print(find_intersection_values(nums1, nums2))


def sum_counts(nums):
    n = len(nums)
    answer = 0

    for i in range(n):
        distinct = set()

        for j in range(i, n):
            distinct.add(nums[j])

            count = len(distinct)
            answer += count * count

    return answer


nums = [1, 2, 1]

print(sum_counts(nums))


def count_pairs(nums, k):
    positions = {}
    answer = 0

    for j in range(len(nums)):

        if nums[j] in positions:
            for i in positions[nums[j]]:
                if (i * j) % k == 0:
                    answer += 1

        if nums[j] not in positions:
            positions[nums[j]] = []

        positions[nums[j]].append(j)

    return answer


nums = [3, 1, 2, 2, 2, 1, 3]
k = 2

print(count_pairs(nums, k))


def find_max(arr):
    maximum = arr[0]

    for x in arr:
        if x > maximum:
            maximum = x

    return maximum


arr1 = [1, 2, 3, 4, 5]
arr2 = [7, 7, 7, 7, 7]
arr3 = [-10, 2, 3, -4, 5]

print(find_max(arr1))
print(find_max(arr2))
print(find_max(arr3))
```

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Python 3.14.0 (v3.14.0:ebf955df7a8, Oct  7 2025, 08:20:14) [Clang 16.0.0 (clang-1600.0.26.6)] on darwin
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: /Users/sherif/Slot D DAA/lab 1.py ==========
ada
>>>
========== RESTART: /Users/sherif/Slot D DAA/lab
ada
[2, 1]
>>>
========== RESTART: /Users/sherif/Slot D DAA/lab
ada
[2, 1]
15
4
5
7
5
>>>
```